# Pratibimb: system architecture

EHR + sparse fingersticks (or CGM) + wearables -> personalised physiological twin -> forecast, risk, and when to test

## 1 DATA IN

### EHR summary

Age, BMI, waist, HbA1c, fasting glucose, regimen (metformin, sulfonylurea)

### Wearable summary

Resting heart rate and daily steps (optional)

### Meal, dose, activity log

Carbohydrate, glycaemic index, food order, drug doses, exercise

### Glucose readings

Fingerstick (SMBG) 1-4 a day, or CGM every 15 minutes

## 2 TWIN CORE

### Prior builder

Soft assignment to the 4 INSPIRED Indian T2D endotypes (SIDD, IROD, CIRDD, MARD) gives a log-normal prior on 6 patient parameters; wearables nudge insulin sensitivity.

*prior over parameters*

### Mechanistic physiology model

Extended Bergman minimal model: 10 states, 6 patient parameters. Gut absorption, sulfonylurea PK/PD, metformin, exercise, circadian rhythm, dawn effect.

*propagate states*

### Data assimilation: particle-filter twin

Joint state + parameter filter, 1,000 particles. Student-t likelihood, Liu-West resampling, slow drift of insulin sensitivity, basal glucose and SU response.

*replay*

### Event-sourced audit log

Every reading, meal and dose is an immutable event; late data triggers deterministic replay.

## 3 DECISIONS AND OUTPUTS

### Forecast with uncertainty

Glucose bands (10-90% and 2.5-97.5%) and risk: P(<70), P(<54), P(>250 mg/dL).

### Active-sensing planner

Picks the reading times with the most information about hypoglycaemia: 'test at 16:30'.

### Twin state vs prior

Insulin sensitivity, basal glucose and SU response with posterior intervals.

### FastAPI service /v1

API-key auth, audit export, erasure endpoint.

**Two-way loop: the twin asks the patient to test at the chosen time; the new reading updates the twin**